

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	1 June 2026
Team ID	
Project Name	ShopEZ: E-Commerce Application
Maximum Marks	4 Marks

Technical Architecture:

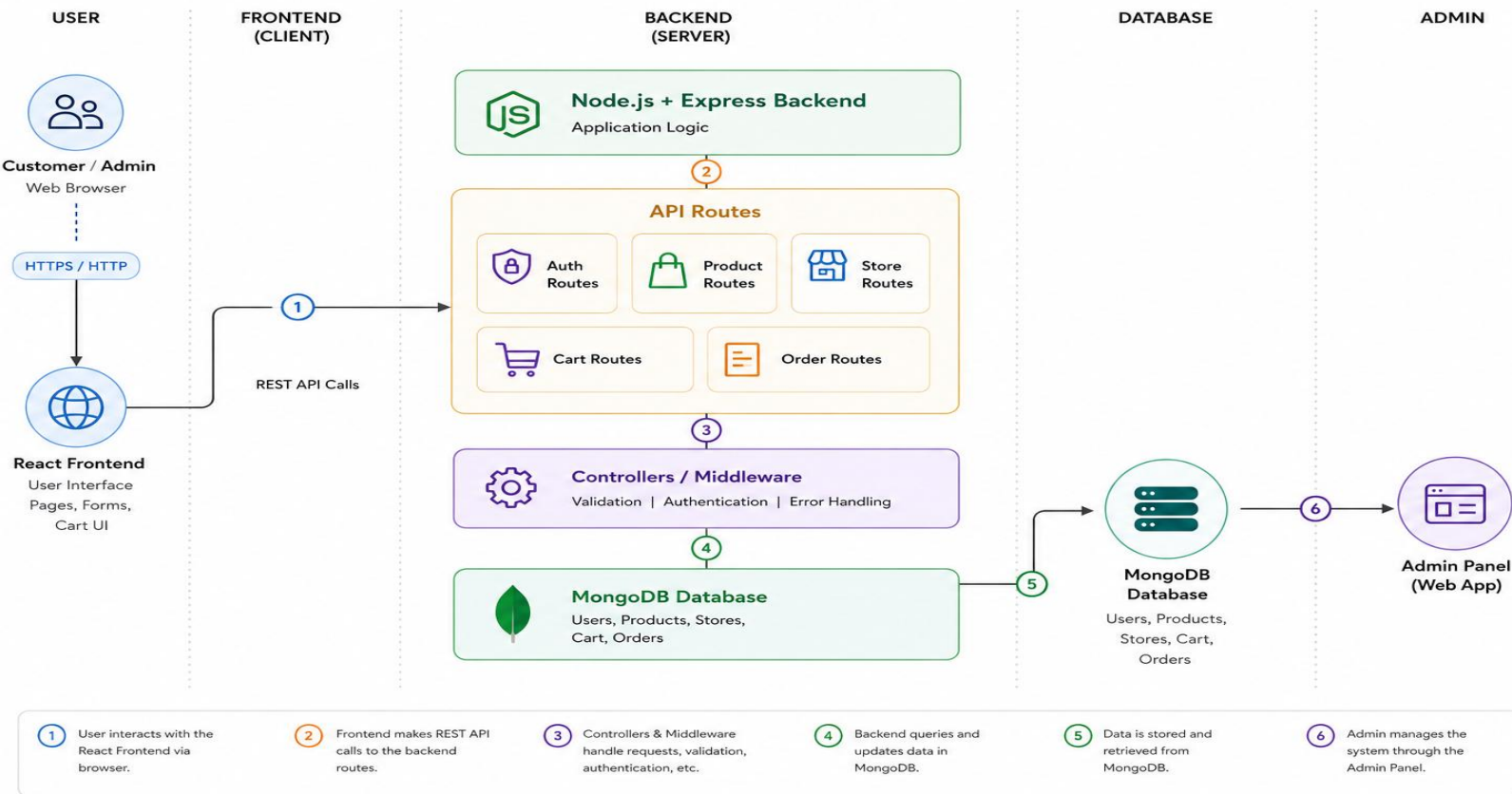


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface used by customers and admins to interact with the application.	React.js, HTML, CSS, JavaScript
2	Application Logic-1	Handles user registration, login, authentication, and user session management.	Node.js, Express.js
3	Application Logic-2	Handles product listing, product details, product creation, update, and deletion.	Node.js, Express.js
4	Application Logic-3	Handles store management, cart management, and order processing.	Node.js, Express.js
5	Database	Stores user, product, store, cart, and order data.	MongoDB / NoSQL
6	Cloud Database	Cloud database service for storing application data if deployed online.	MongoDB Atlas
7	File Storage	Stores product images and static frontend assets.	Local filesystem / Cloud storage, future scope

S.No	Component	Description	Technology
8	External API-1	Payment gateway for online payment processing, if implemented in future.	Razorpay / Stripe / PayPal
9	External API-2	Email or notification service for order confirmation, if implemented in future.	Email API / SMTP, future scope
10	Machine Learning Model	No machine learning model is used in the current version. Can be added later for product recommendation.	Not applicable / Future scope
11	Infrastructure	Application can run locally during development and can be deployed on cloud for production.	Localhost, Render/Railway for backend, Vercel/Netlify for frontend

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The application uses open-source frameworks and libraries for frontend and backend development.	React.js, Node.js, Express.js, MongoDB, Mongoose
2	Security Implementations	User authentication and protected routes are used to secure user/admin access. Passwords should be stored securely using hashing.	JWT, bcrypt, CORS, environment variables, form validation
3	Scalable Architecture	The system follows a 3-tier architecture: frontend, backend, and database. Each layer can be scaled separately.	React frontend, Express REST API, MongoDB
4	Availability	The application can be deployed on cloud platforms so users can access it online. Backend and database services can be hosted separately for better availability.	Vercel/Netlify, Render/Railway, MongoDB Atlas
5	Performance	REST APIs, reusable frontend components, database indexing, and optimized queries improve performance. Static files can be served through CDN in production.	React optimization, Express APIs, MongoDB indexes, CDN